

Project Report

Project Title: Import Data using Transform Maps (Spreadsheet)

Platform: ServiceNow **Team ID:** SWTID-2026-1900

1. INTRODUCTION

1.1 Project Overview

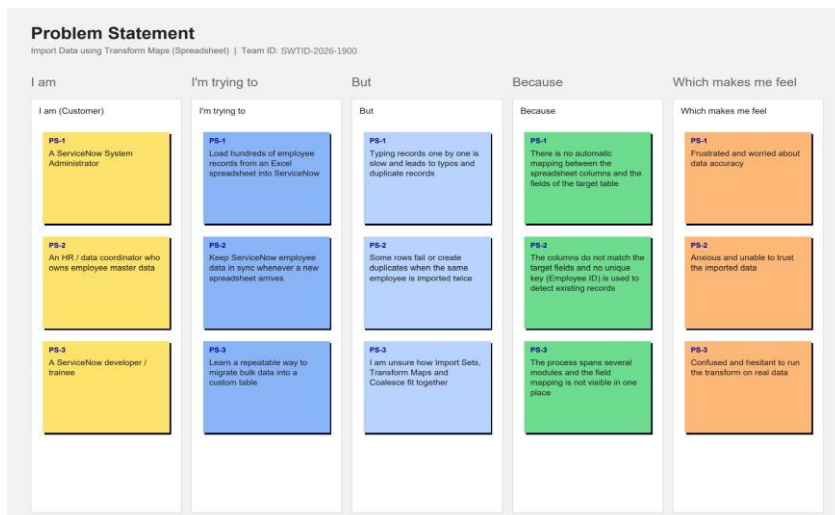
This project imports employee records from a spreadsheet (Excel / CSV) into a custom ServiceNow table. The file is loaded into an Import Set (staging) table, a Transform Map maps the source fields to the target table fields, and the transform is run to create or update records. The imported data is then verified in the target table.

1.2 Purpose

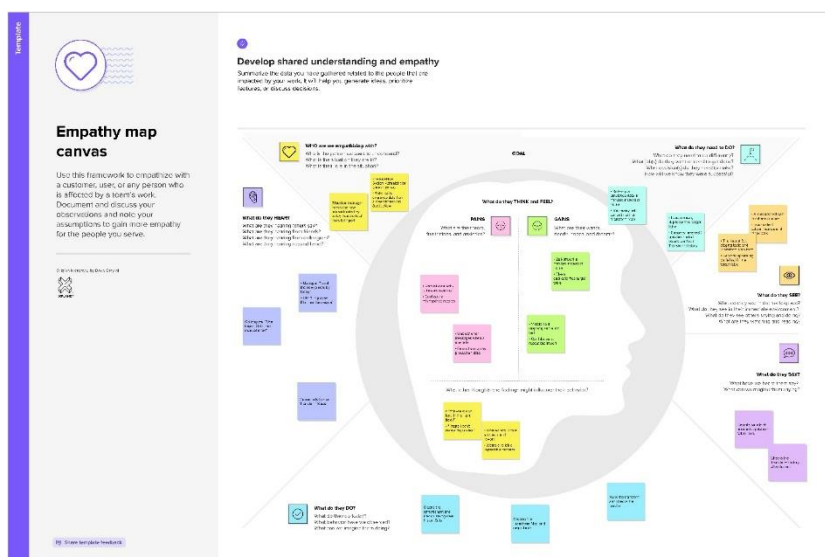
To automate bulk data entry, reduce errors and duplicates, and demonstrate a repeatable data-import process using Import Sets and Transform Maps.

2. IDEATION PHASE

2.1 Problem Statement:



2.2 Empathy mapping canvas:



2.3 Brainstorms:

2 Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

TIP
You can select a sticky note and hit the pencil icon to start drawing!

Vithun.B	Jwelith.K	Manjith Maaran.M	Sanjay.K
Create a custom target table	Use one clean spreadsheet template with fixed column headers	Load file with System Import Sets > Load Data	Run the transform and read the Transform History
Create a Transform Map from staging table to target table	Remove blank rows and duplicate Employee IDs before upload	Create the Import Set (staging) table from the file headers	Compare row counts: file vs staging vs target
Use Employee ID as the Coalesce field to avoid duplicates	Keep a sample dataset of 10 rows for quick testing	Inspect staged rows before transforming	Schedule automated imports in future using a Data Source
Add a transform script for cleaning (trim spaces, lower-case email)			

3 Group ideas

Similar ideas were clustered and each cluster was given a sentence-like label.

20 minutes

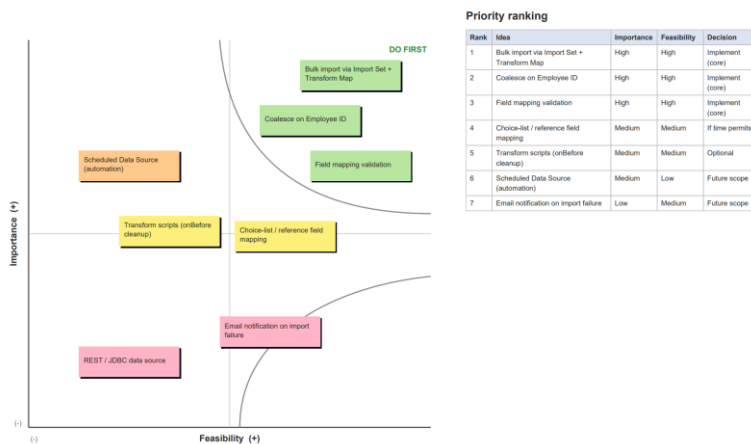
TIP
Tags can be added to sticky notes to find, browse and categorize ideas as themes.

Prepare clean, consistent spreadsheet data	Load the file into an Import Set staging table	Design the Transform Map to avoid duplicates	Validate results and automate later
Fixed template	Load Data module	Target table sample_employee	Transform History
Remove blanks / duplicates	Staging table from headers	Field mapping	Row-count comparison
Sample dataset (10 rows)	Inspect staged rows	Coalesce on Employee ID	Scheduled import (Data Source)
		Transform script (cleanup)	

4 Prioritize

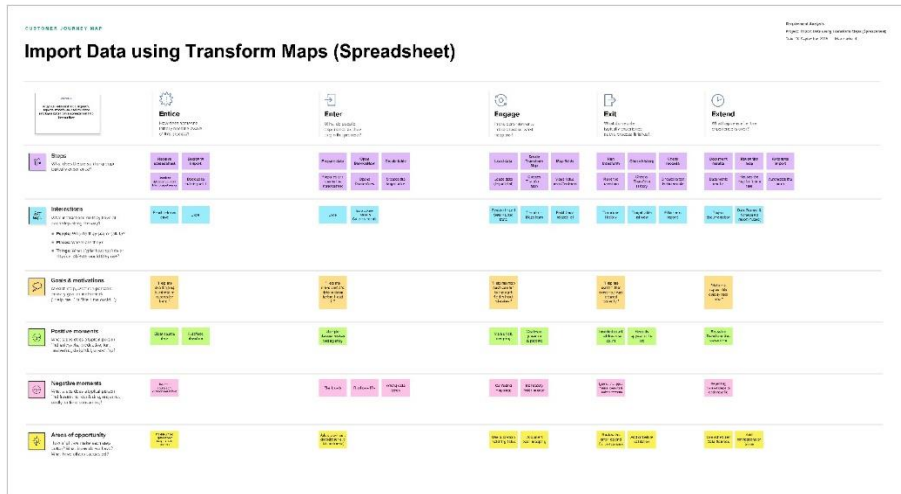
Ideas are placed on the grid by importance (vertical) and Feasibility (horizontal). The top-right is implemented first.

20 minutes



3. REQUIREMENT ANALYSIS

3.1 Customer Journey map:



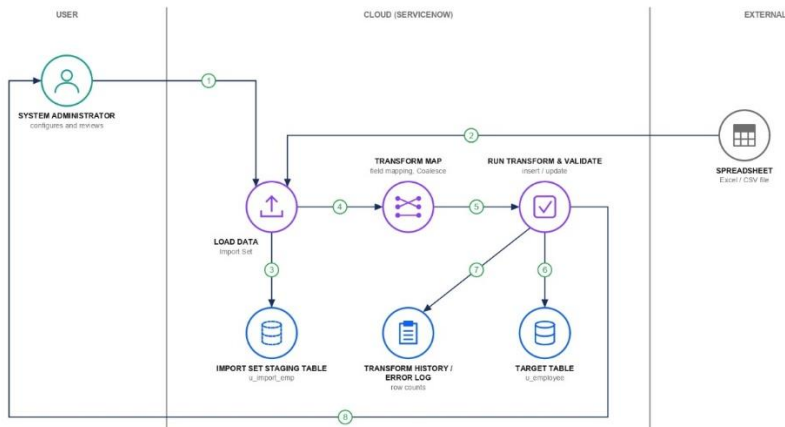
3.2 Solution Requirement:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Preparation	Prepare the employee spreadsheet with columns: Employee ID, Name, Email, Department, Location Clean data: remove blank rows, duplicates and invalid values Save the file as .xlsx or .csv
FR-2	Target Table Setup	Create the custom table (e.g., u_employee) with the required fields Assign appropriate data types (String, Email) and make Employee ID unique / mandatory
FR-3	Import Set Loading	Upload the file using System Import Sets > Load Data Create the Import Set (staging) table from the file headers Review the loaded rows
FR-4	Transform Map Configuration	Create a Transform Map from the Import Set table to the target table Map each source field to its target field (Field Map) Set Employee ID as the Coalesce field
FR-5	Transform Execution	Run the transform from the Transform Map / Import Set View the Transform History
FR-6	Validation & Error Handling	Verify inserted / updated records in the target table Review ignored, skipped or error rows and correct the source data

3.3 Data flow Diagram:

Flow

Data flow diagram: import spreadsheet data into ServiceNow using Transform Maps



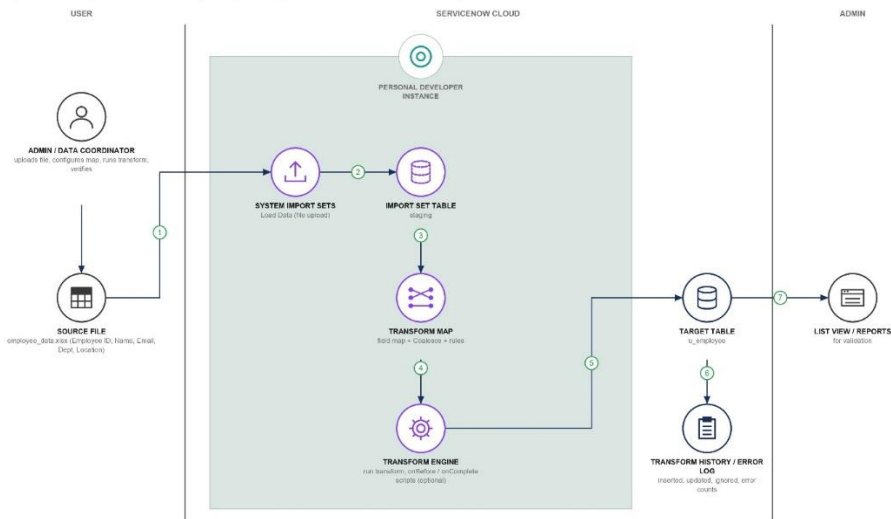
1. Admin selects the spreadsheet file (.xlsx / .csv) to import.
2. Load Data reads the file.
3. Rows are stored in the Import Set Staging Table.
4. Staged rows go to Transform Map (field mapping, Coalesce on Employee ID).
5. Mapped rows go to Run Transform & Validate.
6. Run Transform & Validate inserts / updates records in the Target Table (u_employee).
7. Inserted, updated, ignored and error counts are logged in Transform History / Error Log.
8. Results return to the admin, who reviews them and adjusts the map.

3.4 Technology Stack:

ServiceNow (Personal Developer Instance), System Import Sets (Load Data), Import Set table, Transform Map & Field Maps, Coalesce, Excel / CSV source file.

Technical architecture

Spreadsheet > Import Set > Transform Map > Target Table (ServiceNow Personal Developer Instance)



1. Admin uploads employee_data.xlsx using System Import Sets > Load Data.
2. Rows are loaded into the Import Set Table (staging).
3. Staged rows go to the Transform Map (field map, Coalesce on Employee ID, rules).
4. The Transform Engine runs the transform (onBefore / onComplete scripts optional).
5. The engine inserts / updates records in the Target Table (u_employee).
6. Inserted, updated, ignored and error counts are written to Transform History / Error Log.
7. Admin verifies the data in List View / Reports.

4. PROJECT DESIGN

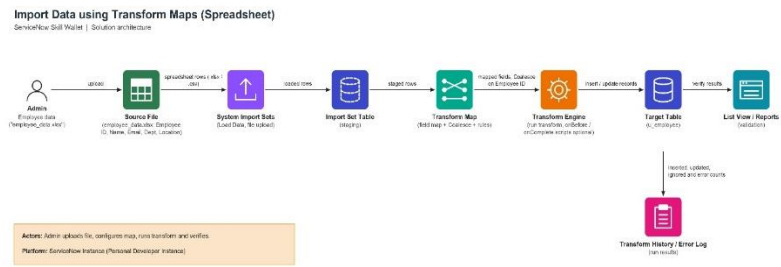
4.1 Problem Solution Fit:

1. CUSTOMER SEGMENT(S) CS ServiceNow system administrators, developers and HR data coordinators who must load employee data from spreadsheets.	8. CUSTOMER CONSTRAINTS CC Limited time, spreadsheet quality varies, must not duplicate or overwrite records, limited access / roles on the instance, PDI may hibernate.	5. AVAILABLE SOLUTIONS AS Manual record entry (slow, error-prone), ad-hoc list import through Excel, custom scripts. Pros: quick for very few records. Cons: no mapping control, duplicates, no audit trail.
2. JOBS-TO-BE-DONE / PROBLEMS J&P Import employee records in bulk into a ServiceNow table accurately and without duplicates.	9. PROBLEM ROOT CAUSE RC Spreadsheet columns do not match table fields; no unique key is used; there is no staging step to review data before it reaches the target table.	7. BEHAVIOUR BE Admins clean the file in Excel, then either type records manually or try list import and fix errors afterwards.
3. TRIGGERS TR A new batch of employees or a data-migration deadline, a manager asks for the data to be available in ServiceNow. 4. EMOTIONS: BEFORE / AFTER EM Before: stressed, insecure about data accuracy. After: confident and in control with a repeatable process.	10. YOUR SOLUTION SL Load the spreadsheet into an Import Set staging table, use a Transform Map with field mappings and Coalesce on Employee ID to insert / update records in the target table, then verify results in the Transform History and list view.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE ServiceNow modules (Load Data, Transform Maps, Transform History), documentation and community forums. 8.2 OFFLINE Team discussions, HR sharing the spreadsheet by email or shared drive, review meetings.

4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Employee data received as spreadsheets has to be entered into ServiceNow manually, which is time-consuming and leads to errors and duplicate records.
2	Idea / Solution description	Import the spreadsheet into a ServiceNow Import Set (staging table) and use a Transform Map to map source fields to the target table fields. Coalesce on Employee ID prevents duplicates. The transform is run and the imported records are validated in the target table.
3	Novelty / Uniqueness	A repeatable, no-code / low-code import that shows the source-to-target mapping in one place, uses a staging table for review before data reaches the target table, and can be reused for future files.
4	Social Impact / Customer Satisfaction	Saves administrators' time, reduces data-entry errors, and gives HR and business users accurate, up-to-date employee data, increasing trust in the platform.
5	Business Model (Revenue Model)	Educational / internal-efficiency project built on ServiceNow. It can be offered as a data-migration service or reusable import template to organizations that use ServiceNow.
6	Scalability of the Solution	The same Transform Map scales to larger files and additional fields; Data Sources and Scheduled Imports (JDBC, REST, file) can automate recurring loads.

4.3 Solution Architecture:

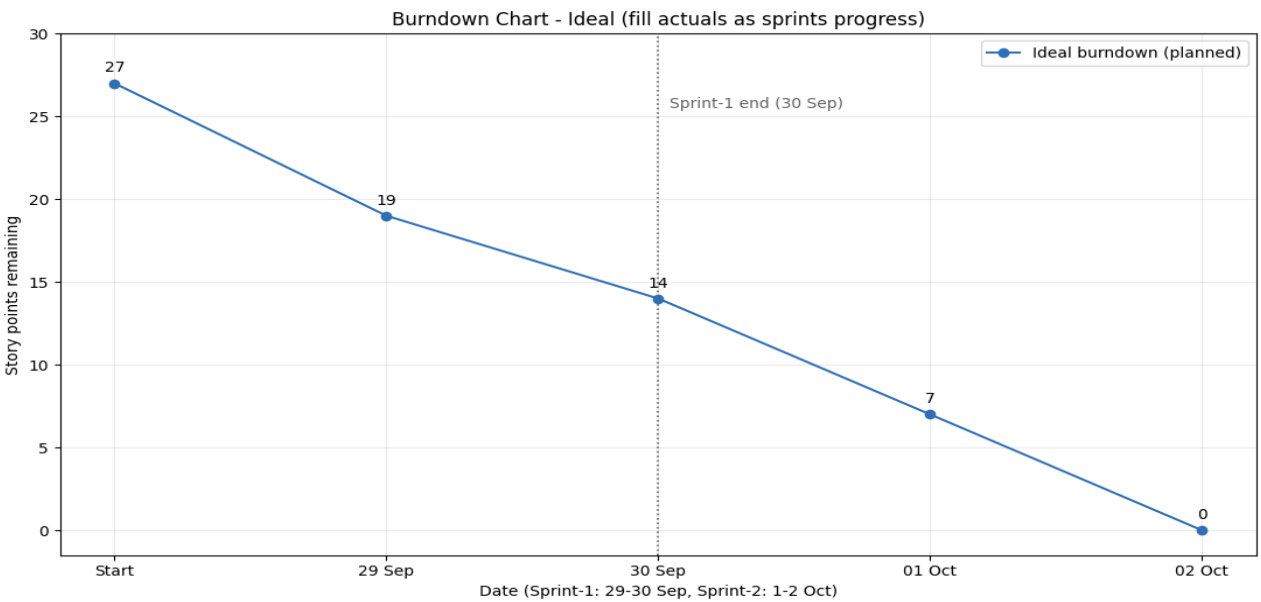


5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

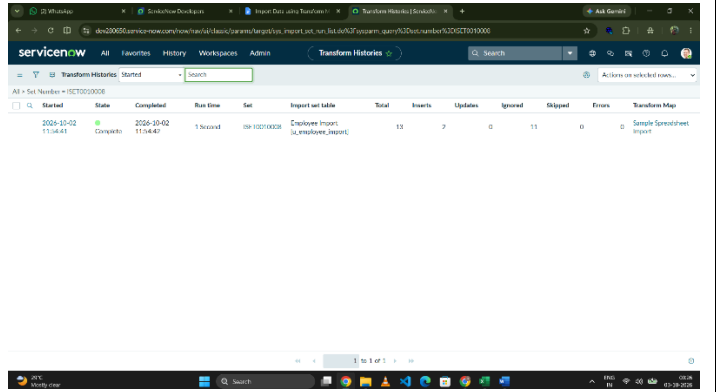
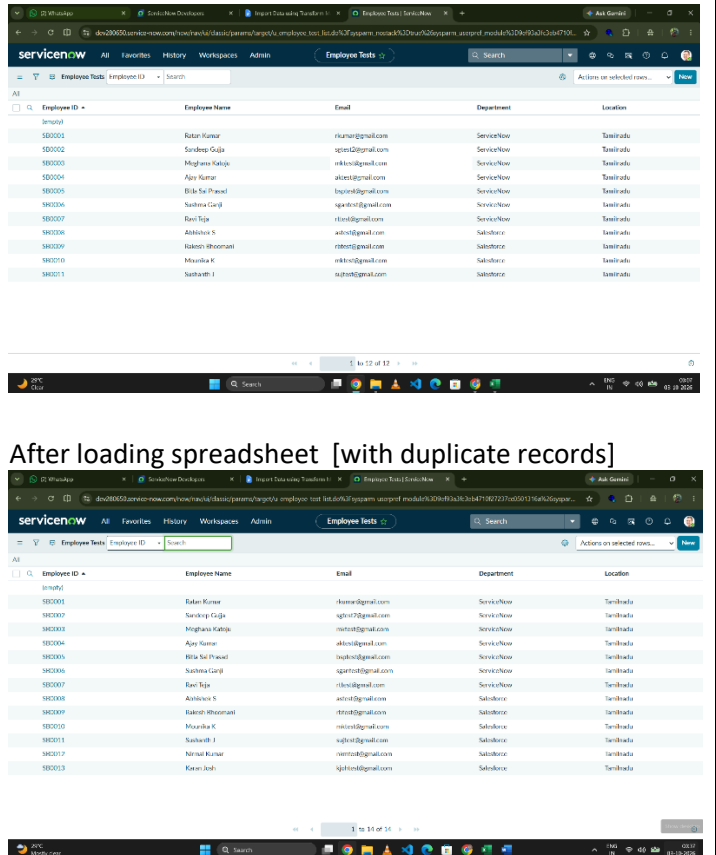
One sprints of 2 days each, 27 story points in total, velocity 13.5 story points per sprint.

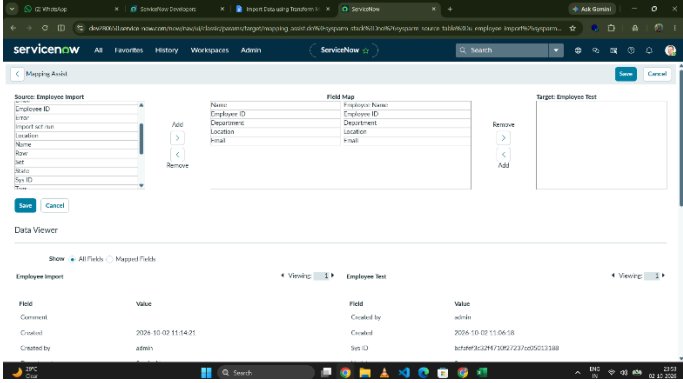
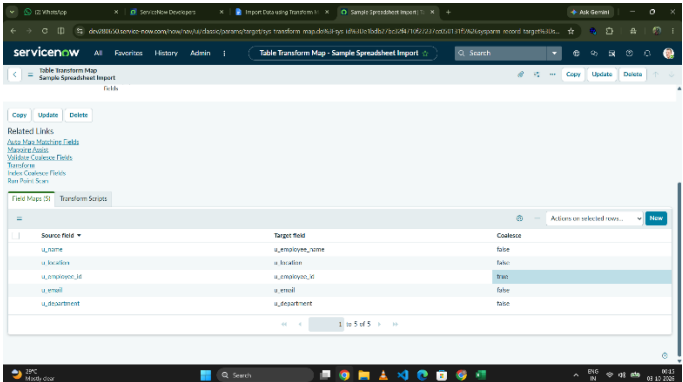
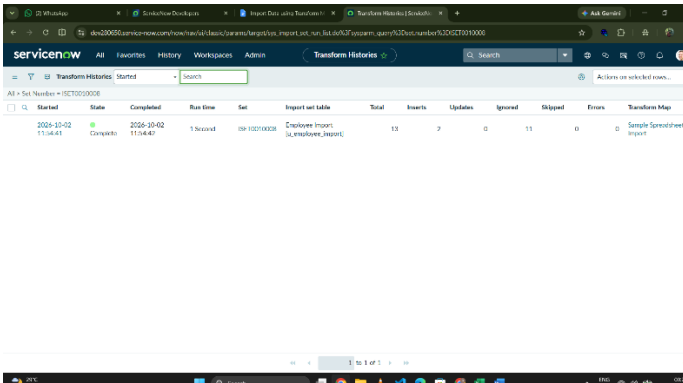
Burnout chat:



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing:

S.No.	Parameter	Values	Screenshot
1	Import Summary	Rows in spreadsheet: 13 Rows loaded into Import Set table: Employee Import[u_employee_import] Inserted: 13 Updated: 2 Ignored: 11 Skipped: 0 Error: 0 Note: If the data already in table was in uploaded spread sheet, it is ignored unless there is any change in the data	
2	Accuracy	Records imported correctly = (Inserted + Updated) ÷ Total rows × 100 = 100% Duplicate records created: 0	

S.No.	Parameter	Values	Screenshot
3	Field Mapping Check	Fields mapped: Employee ID, Name, Email, Department, Location (5 of 5 mapped correctly) Coalesce field: Employee ID	 
4	Transform Time	Load time: 11:54:41 Transform time: 11:54:42 Total Time taken: 1 sec	

7. RESULTS

7.1 Output Screenshots

- Spreadsheet source data
- Load Data screen and Import Set table
- Transform Map and Field Maps
- Coalesce configuration
- Transform History (run result)
- Target table showing imported records

NOTE: ALL THE SCREEN SHOTS ARE KEPT IN A SEPARATE FOLDER AS SUPPORTING SCREEN SHOTS

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Bulk import of records in minutes instead of manual entry
- Field mapping is visible and reusable
- Coalesce prevents duplicate records
- Staging table allows data review before it reaches the target table
- Transform History gives an audit trail

Disadvantages:

- Source data quality directly affects results; it must be cleaned first
- Initial setup of table, mapping and roles needs ServiceNow knowledge
- Large or complex transformations may need scripting
- Manual upload is required unless a scheduled Data Source is configured

9. CONCLUSION

The project shows that spreadsheet data can be imported into ServiceNow accurately and repeatably using Import Sets and Transform Maps, reducing manual effort and data errors.

10. FUTURE SCOPE

- Automate imports with scheduled Data Sources (file, JDBC or REST)
- Add transform scripts for data cleaning and reference-field lookups
- Send notifications when an import has errors
- Import related data such as departments, locations and managers

11. APPENDIX

- Dataset Link: [https://github.com/VithunB/Import-Data-using-Transform-Maps-Spreadsheet-/blob/main/Sample%20sheet%202023%20\(1\).xlsx](https://github.com/VithunB/Import-Data-using-Transform-Maps-Spreadsheet-/blob/main/Sample%20sheet%202023%20(1).xlsx)
- GitHub: <https://github.com/VithunB/Import-Data-using-Transform-Maps-Spreadsheet->
- Demo video URL: <https://drive.google.com/drive/folders/1N6PXR5jH6C3fONm1q5ZEYNSzA-LvFxRT>